

SUPPLY CHAIN ANALYTICS DASHBOARD :

Executive Summary

Efficient supply chain management is essential for organizations to maintain product availability, control operational costs, and deliver products to customers on time. This project presents an interactive **Supply Chain Analytics Dashboard** developed in Power BI to analyze different aspects of supply chain operations, including executive performance, inventory and production, logistics and distribution, and supplier quality.

The dashboard enables stakeholders to monitor important performance indicators such as revenue, production volume, inventory availability, manufacturing costs, shipping costs, lead times, defect rates, and supplier performance. By consolidating these metrics into a single analytical platform, decision-makers can quickly identify operational inefficiencies, compare supplier performance, optimize logistics, and improve overall supply chain efficiency.

Project Objective

The primary objective of this project is to analyze the overall performance of the supply chain and provide actionable insights for business decision-making.

The project aims to:

- Monitor overall supply chain performance using executive-level KPIs.
 - Analyze inventory availability and production efficiency.
 - Evaluate logistics and distribution performance across locations, routes, transportation modes, and shipping carriers.
 - Assess supplier performance based on manufacturing cost, production volume, lead time, and product quality.
 - Identify operational bottlenecks that impact cost, quality, and delivery performance.
 - Support data-driven decisions that improve efficiency, reduce operational costs, and enhance overall supply chain performance.
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Business Problem

Supply chain operations involve multiple interconnected activities, including procurement, production, inventory management, logistics, and quality inspection. Monitoring these

activities separately often makes it difficult for management to identify operational issues and make timely decisions.

Some of the common business challenges include:

- Increasing manufacturing costs.
- Delays in production and product delivery.
- Higher logistics and shipping expenses.
- Variations in supplier performance.
- Product quality issues due to higher defect rates.
- Difficulty identifying the most efficient suppliers.
- Limited visibility into overall supply chain performance.

To address these challenges, organizations require a centralized analytical solution that provides real-time visibility into operational performance and supports informed business decisions.

Dataset Overview

The dataset contains operational information related to various stages of the supply chain process, including production, inventory, logistics, suppliers, manufacturing, customer information, and quality inspection.

The dataset includes information such as:

Product Information

- SKU
- Product Type
- Price

Sales Information

- Products Sold
- Revenue Generated
- Order Quantities

Inventory Information

- Stock Levels
- Product Availability

Supplier Information

- Supplier Name

Production Information

- Production Volume

Manufacturing Information

- Manufacturing Cost
- Manufacturing Lead Time

Logistics Information

- Shipping Costs
- Shipping Time
- Lead Time
- Transportation Modes
- Shipping Carriers
- Routes
- Locations

Quality Information

- Inspection Results
- Defect Rates

This comprehensive dataset enables analysis of operational efficiency, supplier performance, logistics optimization, production effectiveness, and quality management across the entire supply chain.

Data Preparation

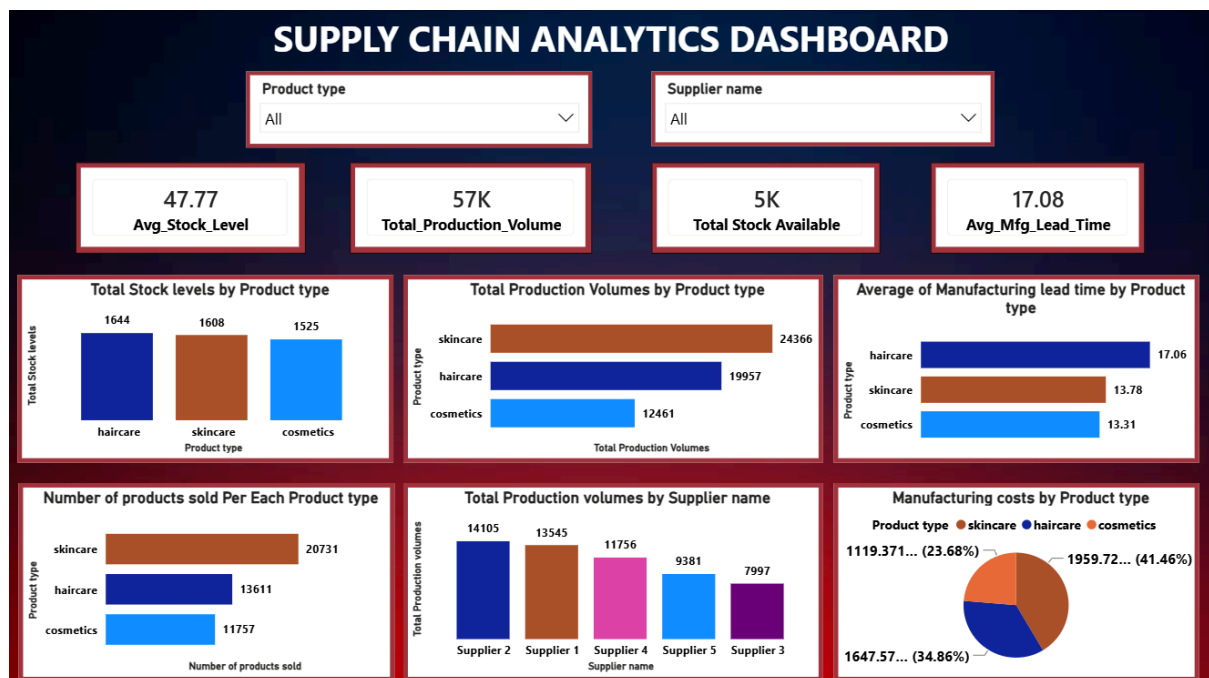
Before developing the dashboards, the dataset was prepared to ensure data consistency and analytical accuracy.

The following preprocessing steps were performed:

- Imported the dataset into Power BI.
- Reviewed the dataset for missing and inconsistent values.
- Verified data types for numerical and categorical columns.
- Validated business-related fields such as production volume, shipping costs, manufacturing costs, and defect rates.
- Created calculated measures using DAX to summarize important business metrics, including average manufacturing cost, average lead time, average shipping time, total production volume, total revenue, and average defect rate.
- Designed interactive slicers to enable dynamic filtering by supplier, product type, location, transportation mode, shipping carrier, and inspection results.

- Organized the dashboard into four business-focused analytical sections:
 - Executive Overview
 - Inventory & Production Analysis
 - Logistics & Distribution Analysis
 - Quality Control & Supplier Performance

➔ Inventory & Production Analysis:



Objective

The objective of this dashboard is to monitor inventory availability, evaluate production performance, and assess manufacturing efficiency across different product categories and suppliers. It helps management ensure adequate stock levels, optimize production planning, and improve manufacturing operations.

Business Questions

- Is the current inventory sufficient to meet customer demand?
- Which product category has the highest stock level?
- Which product category has the highest production volume?
- Which supplier contributes the highest production volume?
- Which product category has the highest number of products sold?

- Which product category requires the longest manufacturing lead time?
 - Which product category has the highest manufacturing cost?
 - Is production aligned with product demand?
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Key Insights

Inventory Performance

The company maintains an average stock level of **47.77**, with a total available stock of **5K** units. Among all product categories, **Haircare** has the highest stock level, followed closely by **Skincare**, while **Cosmetics** has the lowest stock availability.

Production Performance

The total production volume is **57K** units. **Skincare** has the highest production volume (**24,366 units**), followed by **Haircare (19,957 units)** and **Cosmetics (12,461 units)**. This indicates that the company prioritizes skincare production to meet higher market demand.

Product Demand

The sales analysis shows that **Skincare** recorded the highest number of products sold (**20,731 units**), followed by **Haircare** and **Cosmetics**. This suggests that production is largely aligned with customer demand.

Supplier Contribution

Among all suppliers, **Supplier 2** contributes the highest production volume (**14,105 units**), followed closely by **Supplier 1**. This indicates that these suppliers play a major role in supporting the company's production operations.

Manufacturing Efficiency

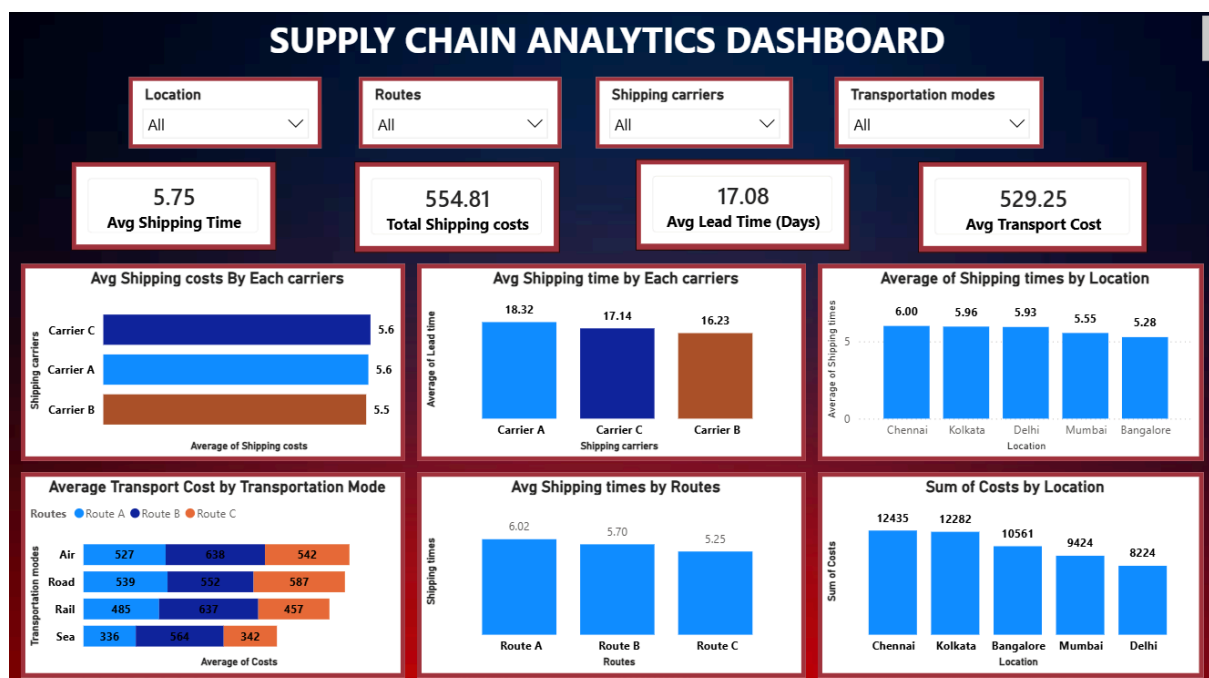
The average manufacturing lead time is **17.08 days**. **Haircare** products require the longest manufacturing lead time, while **Cosmetics** have the shortest. Additionally, **Skincare** has the highest manufacturing cost, indicating that higher production investment is concentrated in the company's highest-performing product category.

Business Recommendations

- Maintain adequate inventory for **Skincare** products to meet consistently high customer demand.

- Review the manufacturing process for **Haircare** products to reduce lead time and improve production efficiency.
- Continue strengthening relationships with **Supplier 2** and **Supplier 1**, as they contribute the highest production volumes.
- Regularly monitor manufacturing costs for **Skincare** products to ensure profitability while maintaining production quality.
- Continuously align production planning with sales demand to minimize stock shortages and excess inventory.

➔ Logistics & Distribution Analysis :



Objective

The objective of this dashboard is to evaluate the efficiency of the company's logistics and distribution operations by analyzing shipping costs, shipping time, transportation costs, delivery routes, shipping carriers, and locations. It helps management identify opportunities to optimize transportation, reduce logistics costs, and improve delivery performance.

Business Questions

- Which shipping carrier has the highest shipping cost?

- Which shipping carrier takes the longest to deliver products?
 - Which transportation mode incurs the highest transportation cost?
 - Which route has the highest shipping time?
 - Which location experiences the longest shipping time?
 - Which location has the highest logistics cost?
 - How efficient are the company's logistics operations?
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Key Insights

Shipping Performance

The average shipping time across all deliveries is **5.75 days**, indicating the overall delivery performance of the supply chain. Among all shipping carriers, **Carrier A** records the highest average shipping time (**18.32 days**), while **Carrier B** has the lowest (**16.23 days**), making it the fastest carrier in the current dataset.

Shipping Cost Analysis

The total shipping cost is **554.81**. The average shipping cost remains relatively similar across all carriers, indicating that shipping expenses are fairly balanced regardless of the carrier selected.

Transportation Cost Analysis

The average transport cost is **529.25**. Among all transportation modes, **Road Transport** records the highest transportation cost, whereas **Sea Transport** has the lowest transportation cost, making it the most economical transportation option.

Route Performance

Among all delivery routes, **Route A** has the highest average shipping time (**6.02 days**), followed by **Route B (5.70 days)**, while **Route C (5.25 days)** performs the best with the shortest delivery time.

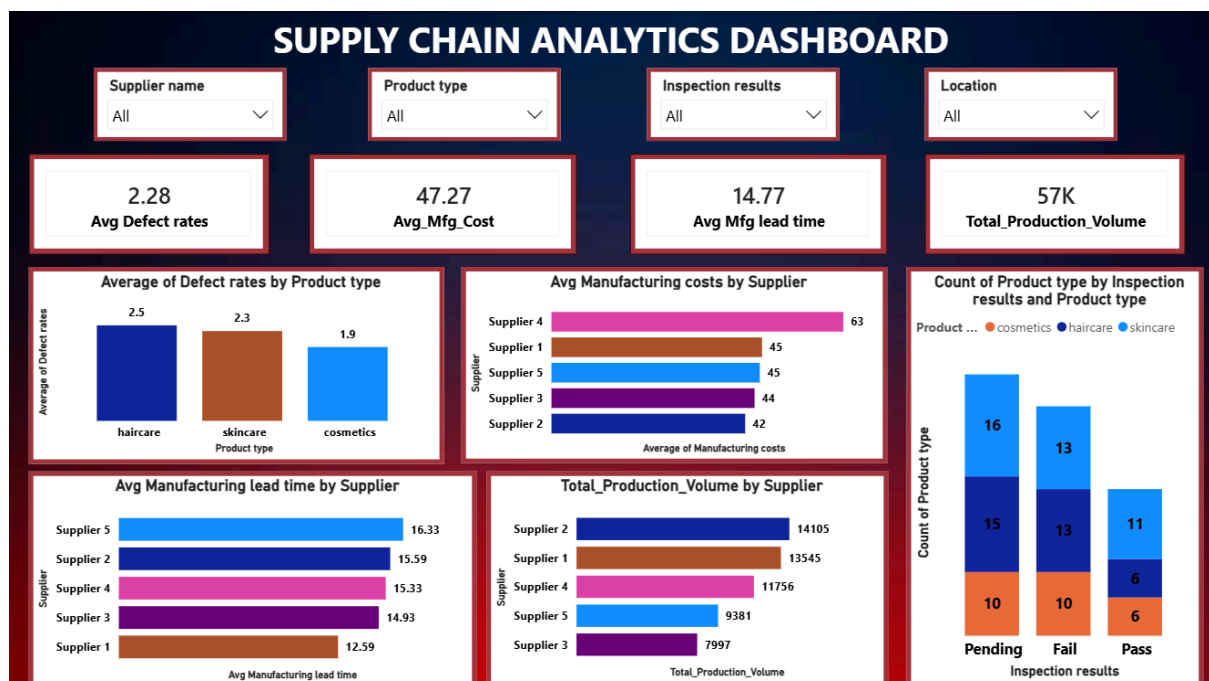
Location Analysis

Chennai records the highest logistics cost (**12,435**) and also has the highest average shipping time (**6.00 days**). In contrast, **Delhi** has the lowest logistics cost (**8,224**) and the shortest average shipping time (**5.28 days**), indicating comparatively better logistics efficiency.

Business Recommendations

- Review logistics operations in **Chennai** to reduce delivery time and transportation costs.
- Analyze **Route A** to identify operational bottlenecks and improve delivery efficiency.
- Evaluate opportunities to increase the use of **Sea Transport** where feasible to optimize transportation costs.
- Monitor the performance of **Carrier A** and identify ways to reduce shipping time.
- Continuously monitor logistics performance across carriers, routes, and locations to improve delivery efficiency and control operational costs.

➔ Quality Control & Supplier Performance:



Objective

The objective of this dashboard is to evaluate supplier performance and product quality by analyzing defect rates, manufacturing costs, production volume, manufacturing lead time, and inspection results. It helps management identify high-performing suppliers, improve product quality, and optimize manufacturing operations.

Business Questions

- Which supplier has the highest production volume?
 - Which supplier has the highest manufacturing cost?
 - Which supplier has the shortest manufacturing lead time?
 - Which product category has the highest defect rate?
 - Which product category has the lowest defect rate?
 - What are the inspection results across different product categories?
 - Which suppliers require quality or process improvements?
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Key Insights

Product Quality

The average defect rate across all products is **2.28**. Among the product categories, **Haircare** has the highest defect rate (**2.5**), followed by **Skincare** (**2.3**), while **Cosmetics** has the lowest defect rate (**1.9**). This indicates that Haircare products require greater attention to quality control.

Supplier Performance

Supplier 2 contributes the highest production volume (**14,105 units**), followed closely by **Supplier 1** (**13,545 units**). In contrast, **Supplier 3** has the lowest production volume (**7,997 units**), indicating lower production contribution.

Manufacturing Cost

The average manufacturing cost is **47.27**. **Supplier 4** has the highest average manufacturing cost (**63**), while **Supplier 2** has the lowest (**42**). This variation suggests differences in production efficiency and cost management among suppliers.

Manufacturing Efficiency

The average manufacturing lead time is **14.77 days**. **Supplier 5** has the longest manufacturing lead time (**16.33 days**), whereas **Supplier 1** has the shortest (**12.59 days**). Suppliers with longer lead times may require process improvements to enhance production efficiency.

Inspection Results

The inspection analysis shows that products fall into **Pending**, **Fail**, and **Pass** categories across all product types. The presence of both pending and failed inspections indicates opportunities to strengthen quality assurance processes before products move further in the supply chain.

Business Recommendations

- Prioritize quality improvement initiatives for the **Haircare** product category to reduce defect rates.
- Review the manufacturing processes of **Supplier 4** to understand and optimize higher manufacturing costs.
- Improve production efficiency for **Supplier 5** by reducing manufacturing lead time.
- Continue strengthening partnerships with **Supplier 2** and **Supplier 1**, as they contribute the highest production volumes.
- Reduce pending inspection cases and investigate failed inspections to improve product quality and ensure consistent manufacturing standards.

Conclusion

The **Supply Chain Analytics Dashboard** provides a comprehensive view of the organization's supply chain operations by integrating executive performance, inventory and production, logistics and distribution, and supplier quality into a single analytical platform.

The analysis revealed that **Skincare** is the highest revenue-generating and highest-producing product category, while **Supplier 1** and **Supplier 2** play significant roles in revenue generation and production. The logistics analysis identified opportunities to optimize shipping costs, transportation modes, and delivery routes, while the quality dashboard highlighted differences in supplier performance, manufacturing costs, lead times, and defect rates that require continuous monitoring.

By consolidating these insights into interactive dashboards, management can monitor key performance indicators, identify operational inefficiencies, optimize inventory and logistics, improve supplier performance, strengthen quality control, and make informed business decisions. Overall, the dashboard serves as an effective decision-support tool for improving operational efficiency, reducing costs, and enhancing overall supply chain performance.

